

Machine learning - Introduction

Regression

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Statistical modeling and Machine learning

- Set of tools for modeling and understanding data.

Statistical modeling: emphasizes statistical inference in low dimensional problems.

Machine learning: emphasizes prediction in high dimensional problems.

Supervised and unsupervised learning

Supervised learning: Finding a model that relates a response variable to a set of predictor variables.

- ▲ Inference
- ▲ Prediction
- ▲ Regression
- ▲ Classification

Unsupervised learning: Finding a hidden structure in a set of data without a response variable.

- ▲ Cluster analysis
- ▲ Principal Component Analysis (PCA)



Inference and Prediction

In many problems one wants to estimate the relation between:

- ▲ input variables
 - explanatory variables*
 - independent variables*
 - covariates*
 - features*
- ▲ output variable
 - outcome*
 - response*
 - dependent variable*
 - label*

$$Y = f(X) + \epsilon \quad X = (X_1, X_2, \dots, X_p)$$

Why estimating f ?

- ▲ Inference
- ▲ Prediction

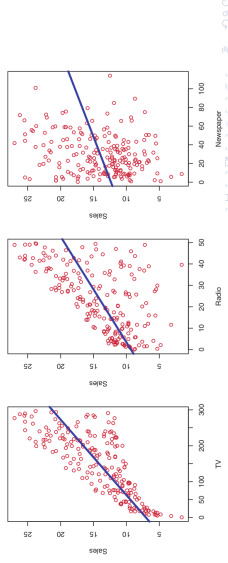


Inference

Aim: understanding how Y is influenced by the X s.

- ▶ Which explanatory variables are important? Which are associated with the outcome?
- ▶ How is the relation between each explanatory variable and the outcome?
- ▶ Positive or negative association
- ▶ Linear or other

Example: Linear regression $Y = \beta_0 + \beta_1 X_1$



Prediction

Aim: predict value of Y given a set of X s.

- ▶ Typically the exact form of f does not matter as long as we get accurate predictions.
- ▶ We want to estimate $\hat{f}$ in a way that minimizes the reducible error.
 - **Training data:** fit/train the model.
 - **Testing data:** test accuracy of the predictions.

The accuracy of the prediction $\hat{Y}$ of Y depends on:

- ▶ reducible error: error in the estimate $\hat{f}$ of f
- ▶ irreducible error: ϵ - there are unmeasured factors that influence the outcome

Prediction - regression

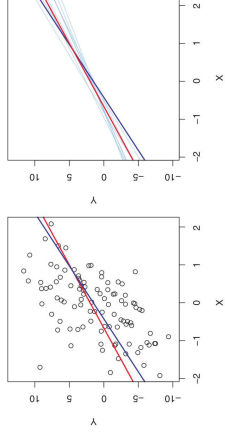


Figure adapted from James, Witten, Hastie, Tibshirani

- ▶ 100 data points simulated from the model $Y = 2 + 3X + \epsilon$
- ▶ The true relation $Y = 2 + 3X$ is shown in **red**
- ▶ ϵ - irreducible error
- ▶ Estimate $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ (blue), can be used to make predictions for new observed values of X .
- ▶ 10 different observations of 100 data points - 10 different $\hat{Y}$
- ▶ For a new observation of X , say x_0 , 10 different predictions $\hat{y}_0$

Quality of fit

There is no single best method for all data sets and questions.

For regression:

Inference

Mean squared error is commonly used:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

- ▶ **MSE (training MSE)** should be small.

Prediction:

- ▶ For a new observation x_0 , $\hat{f}(x_0)$ is close to y_0 .
- ▶ MSE on test data (**test MSE**) should be small.
- ▶ Average $(y_0 - \hat{f}(x_0))^2$ should be small.

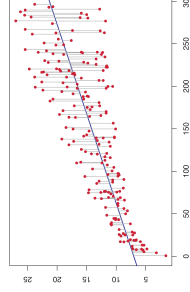
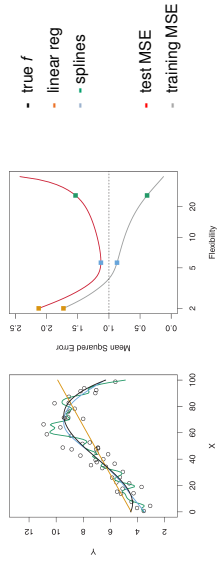


Figure adapted from James, Witten, Hastie, Tibshirani

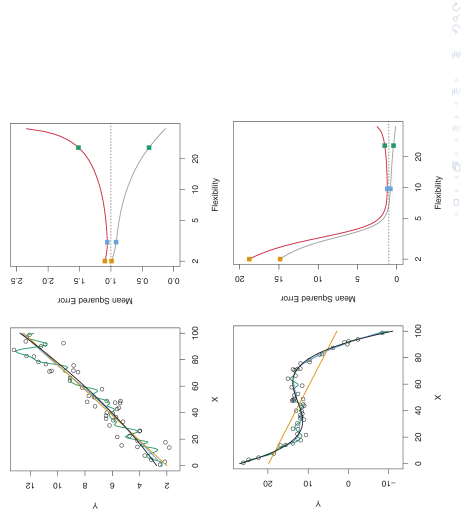
Overfitting

Minimizing training MSE does not imply minimization of test MSE.



Figures adapted from James, Witten, Hastie, Tibshirani

- More flexible models: more accurate on training data, less generalizable for test data.



Overfitting

Linear model, fitting polynomials.

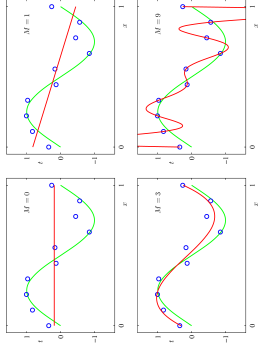
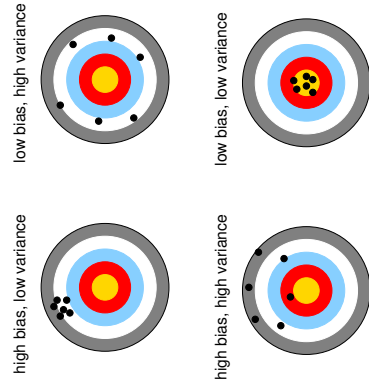


Figure adapted from Bishop

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5 + \beta_6 X^6 + \dots$$

Bias and variance

Adapted from slides of J. Mumford



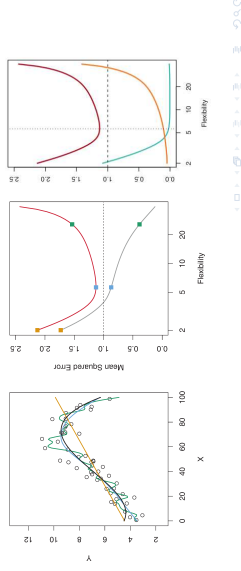
Bias-variance trade off

Variance: amount by which $\hat{f}$ changes when estimated based on other data sets

▲ flexible models $\rightarrow$ more variance

Bias: error introduced by using a given model

▲ flexible models $\rightarrow$ less bias



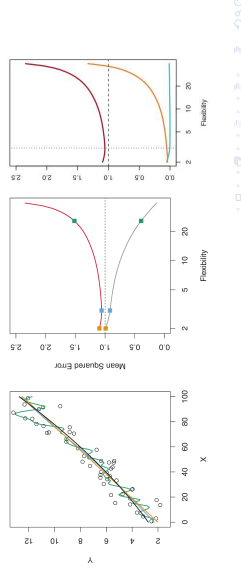
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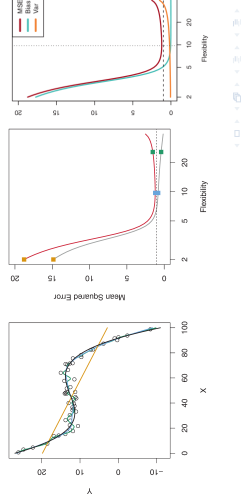
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How to estimate f : parametric approaches

Parametric approaches:

- 1 assume a parametrized form for f
 - 2 estimate the parameters
- Reduces and simplifies the problem of estimating f !

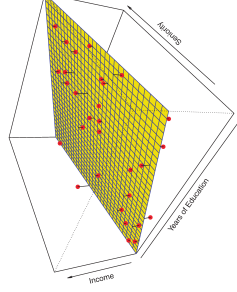


Figure adapted from James, Witten, Hastie, Tibshirani

For example a linear model using least squares for estimation:

- 1 assume $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, $\epsilon \sim N(0, \sigma^2)$
- 2 estimate the β s

How to estimate f : non-parametric approaches

Non-parametric approaches:

- ▶ Do not make explicit assumptions about f , but impose smoothness constraints.
Advantage: can fit a much larger range of f s.
- ▶ The problem of estimating f is not reduced.
Disadvantage: a lot more data is required to estimate f .

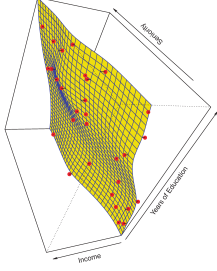


Figure adapted from James, Witten, Hastie, Tibshirani

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Model interpretability versus prediction accuracy

- ▶ What approach for estimating f gives better interpretability?
- ▶ What approach for estimating f gives better prediction accuracy?
- ▶ For what problems does one favor interpretability?
- ▶ For what problems does one favor accuracy?

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K-nearest neighbors regression

64 observations $K = 1$ $K = 9$

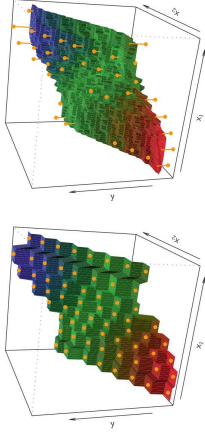


Figure from James, Witten, Hastie, Tibshirani

KNN regression: for a given K and a prediction point x_0 , the estimate of $f(x_0)$ is the average of the K training values closest to x_0 (N'_0).

$$\hat{f} = \frac{1}{K} \sum_{x_i \in N'_0} y_i$$

Figure from James, Witten, Hastie, Tibshirani

Linear regression versus K-nearest neighbors

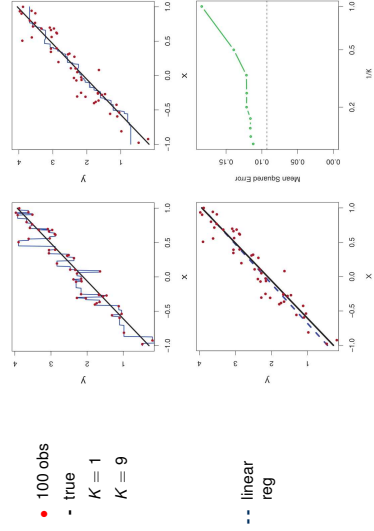


Figure from James, Witten, Hastie, Tibshirani